

Ahmed Hani Abdel Halim El-Gohary

Mechatronics Engineering Student

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PROFESSIONAL SUMMARY

First-Class Mechatronics Engineering student at The British University in Egypt, expected to graduate in 2028. Possesses practical experience in mechanical maintenance, manufacturing operations, engineering design, programming, and industrial automation through multiple internships and technical projects.

EDUCATION

Bachelor of Science in Mechatronics Engineering The British University in Egypt (BUE), Cairo, Egypt

- **Expected Graduation:** 2028
- **Current UK Weighted Average:** 73% (First-Class level)
- **Relevant Coursework:** Thermodynamics, Electric Circuits, Electronics, Mechanical Design, Fluid Mechanics, C Programming, Materials Science, Project Management, Mechanics of Machines.

SKILLS

- **CAD & Design:** SolidWorks, 3D Modelling, Assemblies, Design Visualization, Basic FEA, Motion Simulation.
- **Programming:** C, Python
- **Electronics & Control:** EasyEDA, PCB Design, Basic PLC Knowledge, Sensors and Actuators
- **Engineering Analysis:** Material Selection, Failure Analysis, HVAC Load Estimation, COP and IPLV Evaluation
- **Maintenance & Manufacturing:** Preventive and Corrective Maintenance, Fault Identification, Manufacturing Processes
- **Productivity Tools:** Microsoft Word, Excel, PowerPoint
- **Soft Skills:** Teamwork, Problem Solving, Time Management, Self-Learning, Adaptability.

INTERNSHIP EXPERIENCE

➤ Mechanical Maintenance Intern | Suez Steel Company, Suez, Egypt

Jul 2026 – Aug 2026

- Examined preventive and corrective maintenance procedures for heavy industrial equipment across multiple steel production plants.
- Traced the steel production cycle across melting, casting, and rolling plants, identifying the function and maintenance requirements of key equipment.
- Observed equipment inspection, fault diagnosis, and maintenance activities involving pumps, bearings, gearboxes, conveyors, and hydraulic systems.

➤ Mechanical Maintenance Intern | Modern Drilling Company (MDC), Badr, Egypt

Jan 2026 – Feb 2026

- Gained technical exposure to drilling systems, including top drives, mud pumps, rotary equipment, power transmission, and pressure-control systems.
- Studied the integration of mechanical equipment with electrical drives, sensors, actuators, feedback control, and automation to enable efficient, safe rig operations.

➤ Mechanical Maintenance Intern | Smart Furniture, 10th of Ramadan City, Egypt

Aug 2025

- Gained hands-on exposure to furniture manufacturing, industrial equipment, and mechanical and electrical maintenance practices supporting efficient production.

- Strengthened problem-solving, teamwork, and cross-functional collaboration while developing awareness of quality standards, productivity targets, and workflow discipline.

➤ **Mechanical Maintenance Intern | Ideal Standard, 10th of Ramadan City, Egypt**

Jun 2025 – Jul 2025

- Gained hands-on experience with automated production lines, industrial machinery, sensors, control panels, and maintenance operations in a large-scale manufacturing environment.
- Developed an understanding of PLC-based control, electromechanical troubleshooting, and the integration of mechanical and electrical systems in sanitary ware production.

PROJECTS

○ **Multi-Channel Spiral Extrusion Twist Project | SolidWorks | 2025**

- Designed a complete 3D mechanical system in **SolidWorks**, incorporating a transparent housing for internal flow visualization.
- Performed **FEA stress analysis and motion simulation** to validate structural integrity and mechanical performance.

○ **ROV Frame Material Selection & Failure Analysis | 2025**

- Evaluated Aluminum AA5083, Titanium Ti-6Al-4V, and Stainless Steel 17-4PH for a deep-sea ROV operating at 3,000 m depth.
- Recommended Aluminum AA5083 based on its strength-to-weight ratio, corrosion resistance, manufacturability, and cost-effectiveness.

○ **HVAC Chiller Performance Analysis | 2026**

- Analyzed oil-free centrifugal chillers and estimated hotel cooling loads using **heat-transfer and energy-balance principles**.
- Evaluated full-load and part-load performance under 40°C ambient conditions using **COP and IPLV metrics**.

○ **Maintenance Management System | C Programming | 2026**

- Developed a C-based maintenance management system featuring machine registration, maintenance request tracking, end-of-life monitoring, and an operational dashboard.
- Implemented data structures, date validation, file handling, input validation, and save/load functionality to support reliable equipment records and maintenance scheduling.

○ **12V DC Regulated Power Supply | Electronics Project | 2026**

- Designed and implemented a regulated power supply that converts **220V AC to 12V DC**, incorporating a transformer, bridge rectifier, smoothing capacitors, and an LM7812 voltage regulator.
- Developed the circuit schematic and PCB using **EasyEDA**, then assembled and tested the prototype, achieving a **12.12V** output with a ripple factor below **1%**.

CERTIFICATIONS

• [Fundamentals of Robotics & Industrial Automation](#)

L&T EduTech — Coursera · Issued July 2026

• [Sensors and Sensor Circuit Design](#)

University of Colorado Boulder — Coursera · Issued July 2026

• [Exploring C](#)

University of Michigan — Coursera · Issued May 2026

• [A Quick Tour of Big Data and Business Intelligence](#)

Federica Web learning, University of Naples Federico II — Coursera · Issued June 2026

• [Introduction to Cybersecurity](#)

Almentor · Issued September 2025

• [Leading from the Heart & Managing with Innovation](#)

Almentor · Issued August 2025